

Data Mining for Big Data

Locality Sensitive Hashing

Question 1 : What are the 3-shingles sets in the two following documents?

doc1: "abb abbabb"; doc2: "abb acabb"

What is the Jaccard similarity of doc1 and doc2?

Question 2 : If we consider 26 letters plus the space character, how many different k-shingles are possible (in function of k)?

Question 3 : In a document of n characters, what is the maximum number of different k-shingles?

Question 4 : We construct shingles with words instead of characters. This means that, for instance, a 3-shingle consists of 3 consecutive words in a document (instead of 3 consecutive characters). Consider a document D of n words such that all 3-shingles in this document are different. The document D' is the document D where the last sentence has been moved to the beginning. What is the Jaccard similarity between D and D' ? (You need to compute the number of same/different 3-shingles in this document.)

Question 5 : In the following matrix, each column represents a set (or document) and each row an element (or shingle). There is a "1" in row i and column j if element i belongs to set j . What are the Jaccard similarities between sets A and B, A and C and B and C.

	A	B	C
0	0	1	0
1	0	1	1
2	1	0	1
3	0	0	0
4	1	0	1
5	1	1	1
6	1	0	1

Question 6 : We use the following hash functions to define permutations of rows/elements: $h_1(x) = x$; $h_2(x) = (3x + 1) \bmod 7$; $h_3(x) = (5x + 3) \bmod 7$. Using the row by row (optimized) algorithm, compute the minhash of A and B according to permutations h_1 and h_2 (give the values of minhash at the end of each step for each function and each document).

Question 7 : Compute the minhash of A, B and C according to the 3 permutations. What are the Jaccard similarities inferred from the minhash?

Question 8 : Show that if the Jaccard similarity between two sets X and Y is 0, then the minhash of X and Y is always different (whatever the permutation).

Question 9 : The Jaccard distance is defined as $\text{dist}_{\text{jac}}(A, B) = 1 - \text{Sim}_{\text{jac}}(A, B)$ where Sim_{jac} is the Jaccard similarity. Show that this is a distance, i.e., for all sets A, B and C:

1. it is positive: $\text{dist}_{\text{jac}}(A, B) \geq 0$
2. it is symmetric: $\text{dist}_{\text{jac}}(A, B) = \text{dist}_{\text{jac}}(B, A)$
3. $\text{dist}_{\text{jac}}(A, B) = 0$ if and only if $A = B$
4. it satisfies the triangle inequality: $\text{dist}_{\text{jac}}(A, B) \leq \text{dist}_{\text{jac}}(A, C) + \text{dist}_{\text{jac}}(C, B)$

Question 10 : We want to compute the signature matrix of some sets. The set representation is given as key value pairs (s, i) . This pair means that element i is in set s . Devise a map-reduce framework to compute the signature matrix. You are given n permutation functions p_k . The output signature matrix is represented by pairs $(s, (k, v))$ where s is the set and v is the minhash value of s for the permutation p_k .

Question 11 : Following the previous question, we now want to devise a map-reduce framework to find candidate pairs of sets (k, k') that are similar with high probability. You are given a constant b for the number of bands, r the number of row per band, and the output of the previous question.

Question 12 : Given the family of minhash functions, what is the effect on the probabilities of applying:

1. A 2 way AND construction followed by a 3 way OR construction;
2. A 2 way OR construction followed by a 3 way AND construction;
3. A 2 way OR construction followed by a 2 way AND construction followed by a 2 way OR construction;
4. A 2 way AND construction followed by a 2 way OR construction followed by a 2 way AND construction.

For each case, give also the number of minhash functions used.